MINISTÉRIO DA CIÊNCIA E TECNOLOGIA
INSTITUTO NACIONAL DE PESQUISAS ESPACIAIS

Scientific writing: one paragraph at a time

Research methods course

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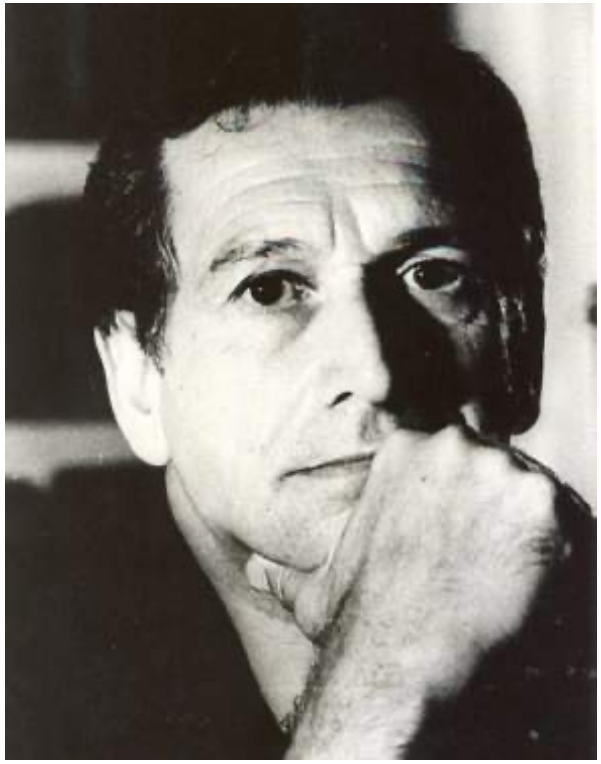
What should you look for?

"In everything in life perfection is finally attained, not when there is nothing else to add, but when there is nothing left to take away."
(Antoine de Saint-Exupéry)

Does writing have to be difficult?

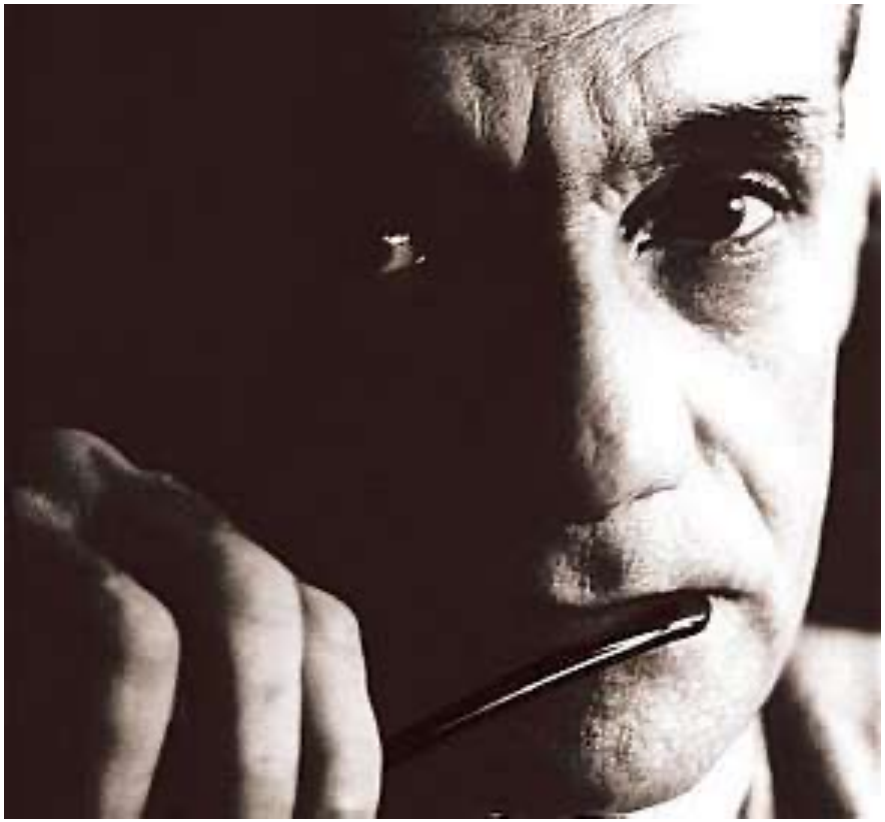


"For me, the act of writing is very difficult and painful, I always have to correct and rewrite several times. Suffice it to say, as an example, that I wrote 1100 typewritten pages to make a novel in which I used just over 300". (Fernando Sabino)

"Make no mistake, I can only achieve simplicity through hard work". (Clarisse Lispector)

Lessons from João Cabral de Melo Neto

"My struggle is to try to put complex things in simple language. In most cases, I fail."



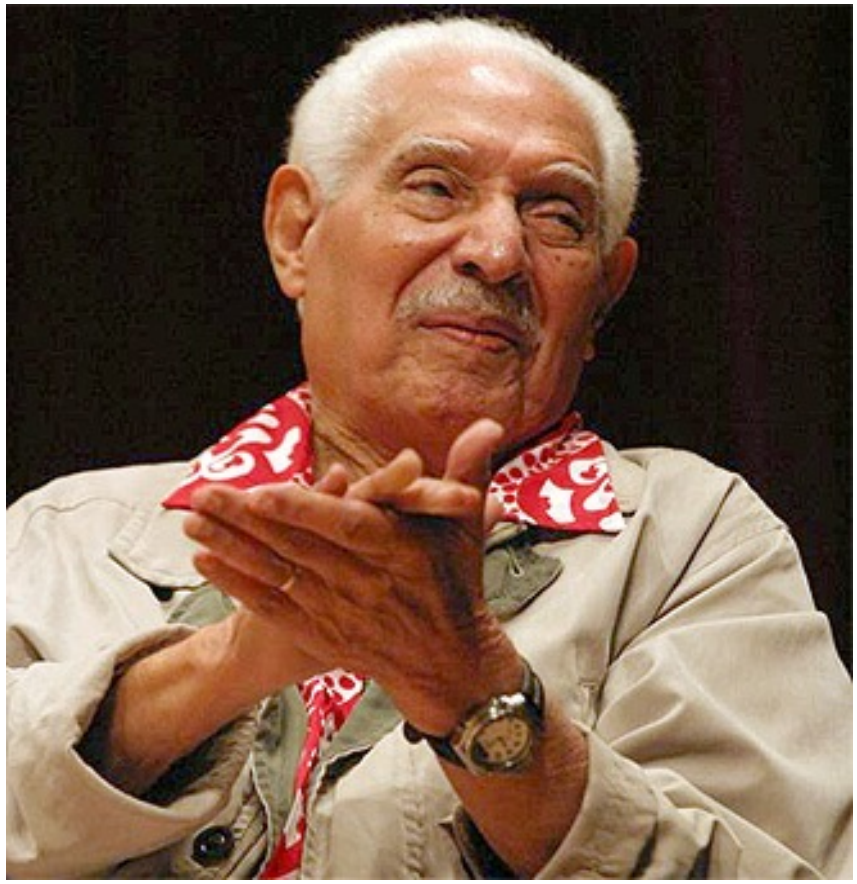
"The words stone or knife or apple, concrete words, are much stronger, poetically, than sadness, melancholy or longing."

Lições de Dorival Caymmi

Só louco
Amou como eu amei.
Só louco
Quis o bem que eu quis.

Ah, insensato coração
Porque me fizeste sofrer?
Porque de amor p'ra entender
É preciso amar?

Por que?
Só louco.



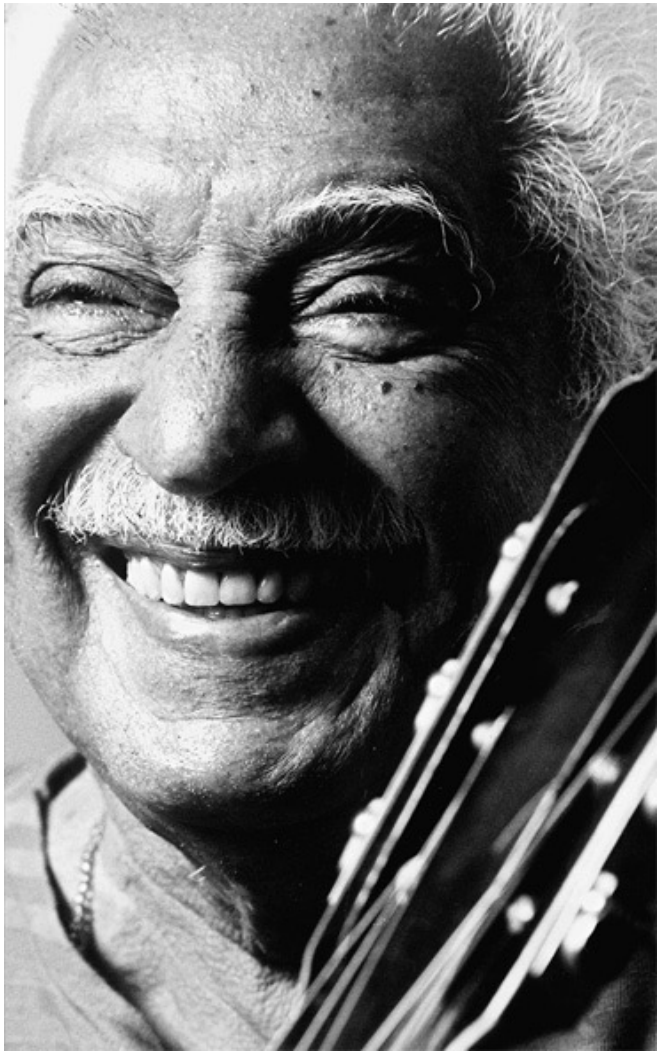
Lições de Dorival Caymmi

Maurino, Dadá e Zeca, ô, Embarcaram de manhã.
Era quarta-feira santa, dia de pescar e de pescador.
Quarta-feira santa, dia de pesca-dor.

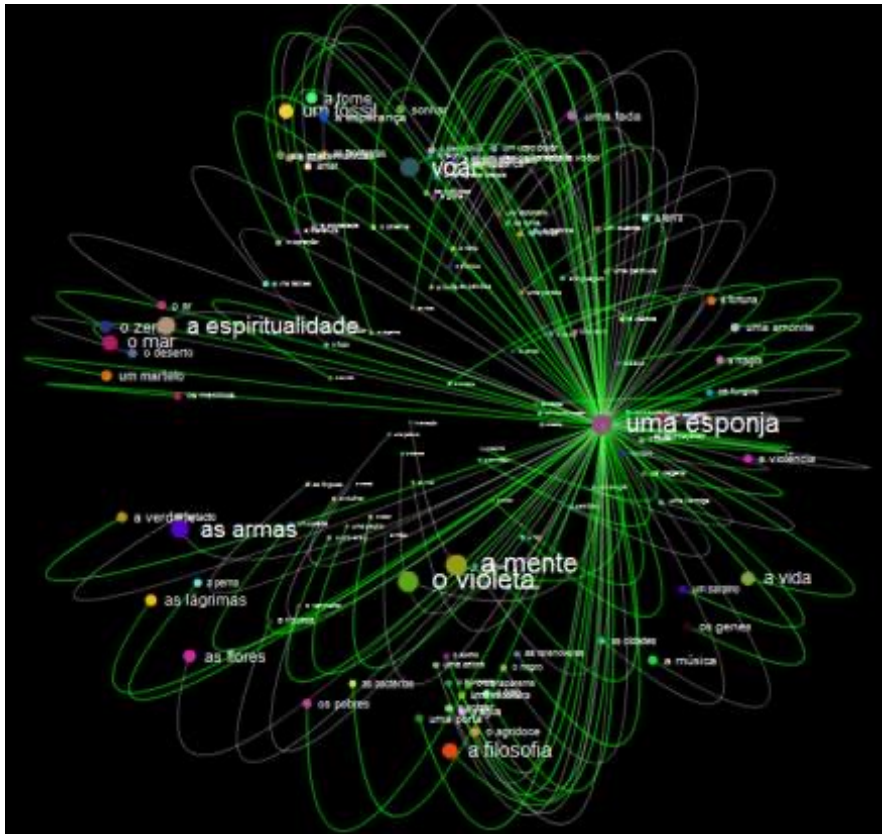
Se sabe que muda o tempo,
Sabe que o tempo vira,
Aí o tempo virou.

Maurino que é de guentar, guentou.
Dadá que é labutar, labutou.
Zeca, esse, nem falou.

Era só jogar a rede e puxar a rede...



Writing is cutting words: microstories



"The woman I loved turns into a ghost. I am the place of apparitions" (José Arreola)

"For sale: unused baby shoes" (Ernst Hemingway).

Lessons from George Orwell

Never use a metaphor, simile, or other figure of speech which you are used to seeing in print.

Never use a long word where a short one will do.

If it is possible to cut a word out, always cut it out.

Never use the passive where you can use the active.

Never use a foreign phrase, a scientific word, or a jargon word if you can think of an everyday English equivalent.

Break any of these rules sooner than say anything outright barbarous.

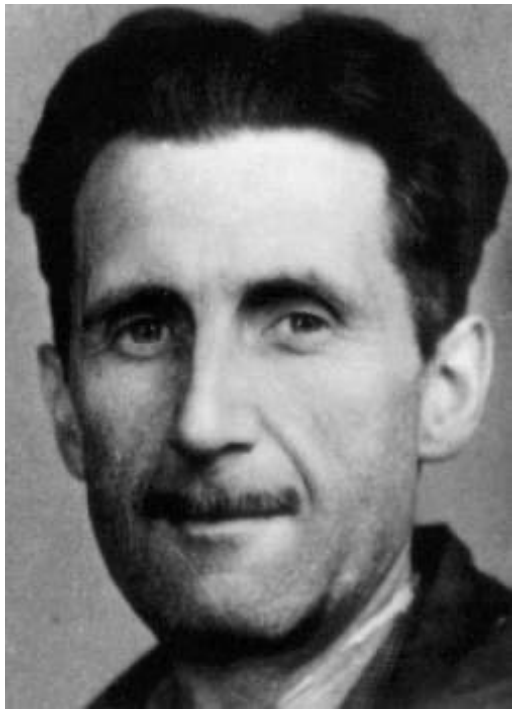


photo source: Wikipedia

Why is one sentence better than another?

Our lack of knowledge about local conditions precluded determination of committee action effectiveness in fund allocation to those areas in greatest need of assistance.

Because we knew nothing about local conditions, we could not determine how effectively the committee had allocated funds to areas that most needed assistance.

Example from W. Booth, G. Colomb, J. Williams, “The Craft of Research”.

Verbs x nouns

Nominalizations: nouns derived from verbs

discover	discovery
move	movement
resist	resistance
react	reaction
fail	failure
refuse	refusal
think	thought

Avoid nominalizations. Always use the original verbs.

Verbs and simple words where appropriate

This demonstrates an understanding of the history, diversity and commonality of the peoples of the nation, the reality of human interdependence, the need for global cooperation, and a multicultural perspective.

Example from W. Booth, G. Colomb, J. Williams, “The Craft of Research”.

Verbs and simple words where appropriate

This demonstrates an understanding of the history, diversity and commonality of the peoples of the nation, the reality of human interdependence, the need for global cooperation, and a multicultural perspective.

This shows we understand why history and common beliefs in each nation matter. Across the world, people need to work together and respect their different cultural backgrounds.

Example from W. Booth, G. Colomb, J. Williams, “The Craft of Research”.

The introduction should impact the reader

The key proposition should be clearly described in the abstract

"This paper develops a spatial statistical analysis of the determinants associated to land-use change in Amazonia. We use a spatially explicit database with 25km × 25km regular cells covering the forest area, including 50 environmental and socio-economic variables to support a spatially explicit statistical analysis".

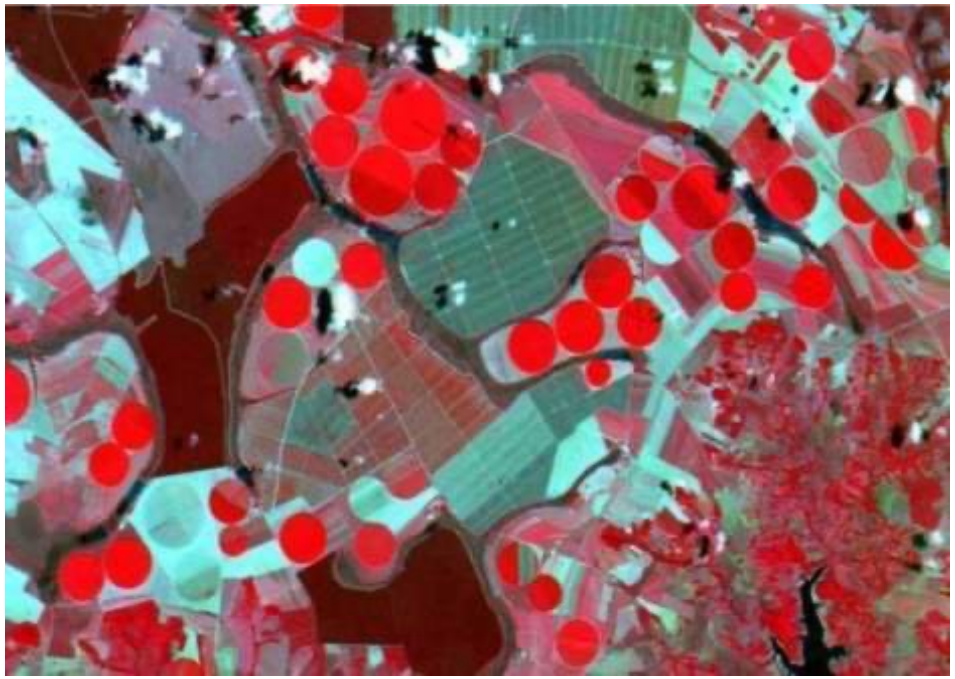
The introduction should impact the reader

The key proposition should be well written and highlighted in the introduction

"Deforestation in the Brazilian Amazonia has a significant impact on biodiversity loss and carbon emission. This paper examines the driving forces behind land change in Amazonia. We performed a spatial statistical analysis, using 50 environmental and socio-economic variables in a database of 25km × 25km cells covering the forest area".

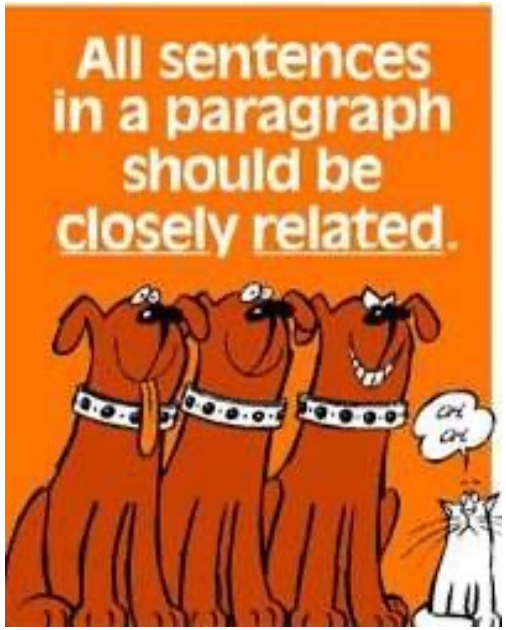
Key proposition

Despite over 30 years of experience in data gathering, processing and analysis and the widespread nature of remote sensing imagery, the ontological status of images remains an open issue. It is surprisingly difficult to provide a straightforward answer to a basic question, “What’s in an image?”



Paragraph-based writing

The paragraph is the basic unit of the scientific text. Each paragraph should contain a single idea.



Paragraph

Topic → Support material → Conclusion

Topic	presents the subject
Support material	develops the ideas
Conclusion	completes the argument.

Paragraph: old before new...

Paragraph

Topic (old)
links with previous paragraphs

Support material
(makes logical sequence)

Conclusion (new)
contains key message
links with next paragraph

Paragraph: old before new...

In previous studies, Smith et al (1999) approached the problem of social exclusion from the point of view of .... Their results do not include.... In our work, the approach adopted allows....

Consistent paragraphs make a difference

"This paper proposes a methodology for mining patterns of change that supports the extraction of spatial configurations from remote sensing image databases. This methodology addresses the problem of describing land use change based on remote sensing image classification. The results show that pattern classification techniques associated to remote sensing image interpretation are a step forwards in the understanding and modeling of land use change. Using satellite images and concepts of landscape ecology, the methodology provides a way to identify deforestation patterns in a complex domain such as Amazon tropical forest."

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"This paper proposes a method for discriminating land change patches extracted from remote sensing image databases and hence describing agents of land use change. The method combines techniques from digital image processing, landscape ecology and data mining. After each image is classified, we measure land change patches using landscape ecology metrics, and then use data mining to label them. We successfully applied the technique for identifying patterns of change in the Amazon tropical forest."

Consistent paragraphs make a difference

"This paper proposes a methodology for mining patterns of change that supports the extraction of spatial configurations from remote sensing image databases. This methodology addresses the problem of describing land use change based on remote sensing image classification. The results show that pattern classification techniques associated to remote sensing image interpretation are a step forwards in the understanding and modeling of land use change. Using satellite images and concepts of landscape ecology, the methodology provides a way to identify deforestation patterns in a complex domain such as Amazon tropical forest."

"Deforestation in Amazonia is done by different agents, including large crop producers, timber extractors, cattle ranchers and settlers. In this paper, we propose a method for describing agents of land use change, using land change patches extracted from remote sensing images. The method combines techniques from digital image processing, landscape ecology and data mining. We successfully applied the technique for identifying agents of change in the Amazon tropical forest.

Don't be a mosquito: don't bite the reader!

Models of Aedes aegypti population dynamics can be classified as deterministic [Ferreira and Yang 2003] or stochastic [Otero et al. 2006; Focks et al. 1993a]. To understand the spatial-temporal dynamics of these populations, this work proposes a new approach to couple Aedes aegypti population dynamic models with local scale spatially-explicit models, which are integrated with geographical databases. This way, at each simulation time step, the variation in population size calculated by the dynamic models are allocated in a grid of rectangular and regular cells used to represent the Geographical Space.

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Studying Aedes aegypti mosquitos is important for public health, as they transmit dengue fever. To understand how Aedes aegypti spread dengue, we propose a model that simulates how mosquitos grow and move in space and time. We calculate the number of mosquitos using a population model and we move the mosquitos in a cell space using a spatial diffusion model. In this way, the model captures how the mosquito spreads dengue.

What's wrong with this paragraph?

Land use changes are the result of the complex interaction between human and biophysical driving forces, which act over a wide range of temporal and spatial scales. The scale on which the process is studied affects the explanation of the phenomenon. Relationships between the land use and the forces driven in local studies cannot be inferred directly for regional scales, due to properties as non-linearity, emergence and collective behavior. In different scales, different processes can have dominant influence on the land use system. Regional dynamics impact and are impacted by local conditions in interactions top-down and bottom-up. According to Becker (2005): “it is impossible today, more than ever, to understand what happens in one place, and consequently, to conceive and implement adequate public policies, without considering the interests and conflicting actions at different geographical scales”.

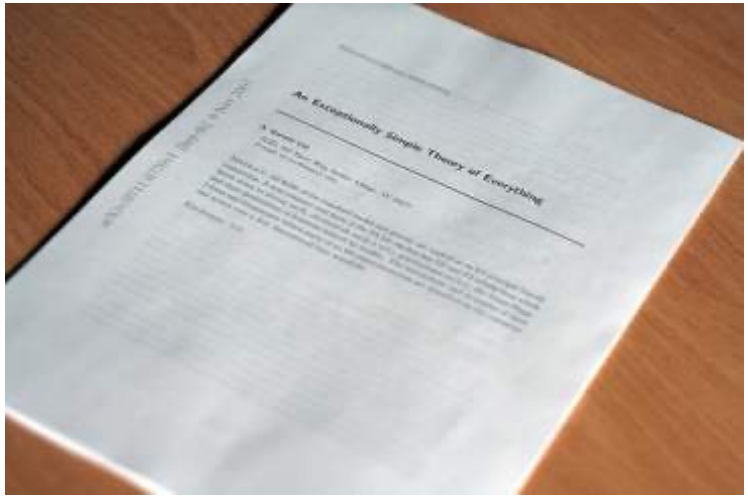
Use of the passive voice
Lack of logical sequence of arguments
Conclusion does not conclude
Paragraph does not breathe!

Improved?

Land change models are useful to foresee trajectories of land use that might result from alternative decisions and policies. When building land use models, we need to capture how humans interact with nature. We can look at the problem at different temporal and spatial scales. Whether we consider society as households, communities, nations, or global companies makes a difference on what we model. To build land change scenarios and support proper public policies, we need to consider driving forces and conflicts at different geographic scales (Becker 2005).

First sentence contains the topic
Arguments follow logic
Conclusion contains the main msg

Self-plagiarism



articles

A physical map of the mouse genome

Reuse of up to 30% of a previous scientific article is the maximum acceptable limit for a new article

Conference article can be used for magazine

Parts of a scientific article can also be reused for a dissemination article

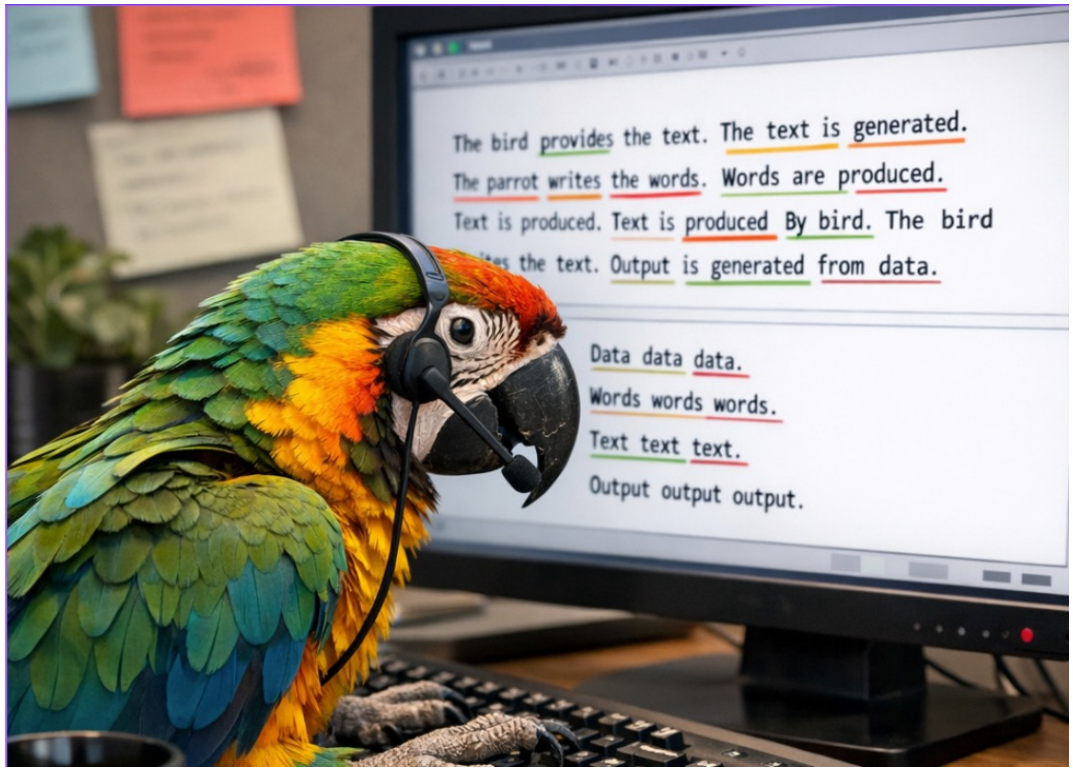
(Pamela Samuelson, Self-Plagiarism or Fair Use, Communications of the ACM, August 1994)

Using LLMs to help your writing

- LLMs are good for:
  - Reviewing the literature (if you already know it)
  - Improving drafts (when you have one)
  - Developing simple code

- LLMs have problems:
  - Text is verbose and boring
  - Uncritical reviews of existing papers
  - Literature surveys can miss important work
  - Sensitive to frequency, not to quality

Avoid being a stochastic parrot



Paragraphs with long sentences
Many sentences with the subject
Unecessary complex words
Marketing language
Not only A... but also B

Image: MS-Copilot

Wikipedia rules to detect LLMs: undue emphasis

Words to watch: stands/serves as a testament/reminder…
a vital/significant/crucial/pivotal/contribution…
underscores/highlights its importance/significance…
setting the stage for… represents/marks a shift…
key turning point… evolving landscape… focal point…

Wikipedia rules to detect LLMs: superficial terms and marketing slogans

....highlighting/underscoring/emphasizing ..., ensuring ..., reflecting/symbolizing ..., contributing to ..., cultivating/fostering ..., encompassing ..., valuable insights, align/resonate with

profound... enhancing .... showcasing ... commitment to... substantial advancement... breakthrough... great novelty, ... groundbreaking... renowned... featuring ...

Avoid being a stochastic parrot

Gemini, write a paragraph on what is data-driven AI

Data-driven AI—which encompasses machine learning and deep learning—functions on a bottom-up paradigm based on statistical pattern recognition rather than explicit human programming. Instead of being given the rules to solve a problem, the system is fed vast amounts of data and uses algorithms to iteratively learn and infer the underlying patterns on its own. This approach excels in complex, fluid environments where rules are practically impossible to hard-code, powering modern breakthroughs in natural language processing, facial recognition, and autonomous driving. While highly adaptable, data-driven AI requires massive computational resources and often operates as a "black box," making it incredibly difficult to explain exactly how or why the system arrived at a particular decision.